The empirical recurrence for OEIS A279980, recovered from its tail

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Abstract

OEIS A279980 records "Empirical recurrence of order 68" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 612$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 11$, so the recurrence is proved for every $n \geq 79$. Its last coefficient is $c_{68} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-68 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A279980 is "Number of 4Xn 0..1 arrays with no element equal to a strict majority of its horizontal and antidiagonal neighbors, with the exception of exactly two elements, and with new values introduced in order 0 sequentially upwards.". It has offset 1 and begins

$$0, 31, 296, 1922, 10491, 50690, 226771, 963728, \ \ldots$$

The entry states:

> Empirical recurrence of order 68 (see link above)

As of the "Last modified" line on the live entry (Dec 24 2016 08:39:04, revision 5) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 612$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 612$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 68: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 11$ is the first whose minimal order is exactly the stated 68.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1224$ exact terms from $n = 11$ on, it returns order 68 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{68} c_i\, a(n-i),$$

c_1	23	c_{18}	497206	c_{35}	−6569381	c_{52}	55428
c_2	−241	c_{19}	−4386716	c_{36}	−27810707	c_{53}	48102
c_3	1516	c_{20}	8790550	c_{37}	41631940	c_{54}	18371
c_4	−6302	c_{21}	−7165006	c_{38}	−8451239	c_{55}	26003
c_5	17644	c_{22}	−2430565	c_{39}	−25767441	c_{56}	1109
c_6	−30647	c_{23}	11152991	c_{40}	22381647	c_{57}	2232
c_7	16375	c_{24}	−9399945	c_{41}	−3872238	c_{58}	−2628
c_8	76525	c_{25}	2868264	c_{42}	−1068125	c_{59}	−2442
c_9	−258610	c_{26}	−6539946	c_{43}	484249	c_{60}	−1234
c_{10}	392963	c_{27}	17657292	c_{44}	−3753572	c_{61}	−1018
c_{11}	−205819	c_{28}	−11665314	c_{45}	2665423	c_{62}	−358
c_{12}	−425216	c_{29}	−19465380	c_{46}	218761	c_{63}	−214
c_{13}	1123927	c_{30}	42565598	c_{47}	578602	c_{64}	−73
c_{14}	−1224977	c_{31}	−25274920	c_{48}	−73694	c_{65}	−25
c_{15}	653452	c_{32}	−10004458	c_{49}	−626246	c_{66}	−10
c_{16}	−262790	c_{33}	14788234	c_{50}	19132	c_{67}	−1
c_{17}	431470	c_{34}	7702753	c_{51}	−120282	c_{68}	−1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 79$, and it is the recurrence OEIS A279980 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 11$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{68} - c_1 x^{68-1} - \cdots - c_{68}$. Its constant term is $-c_{68} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 68 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 68, so $\deg q = 68$, and it is monic. It holds from some index t on. From $\max(t, 11)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 68, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 22 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 68, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-68 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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